

RAM PRAKHYATH ANNAMAREDDY

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OBJECTIVE

Computer Science student with a strong interest in building scalable and efficient applications, supported by hands-on experience in backend development, systems and AI-integrated applications.

EDUCATION

PES University, Bangalore, India [2023-Present]

BTech. – Computer Science & Engineering (CGPA : 7.90)

Relevant Coursework :

- Object Oriented Analysis and Design (Java)
- Cloud Computing
- Database Management Systems
- Operating Systems
- Machine Learning
- Computer Networks
- Data Structures and Algorithms

Certifications :

- Next.js 13: Your Odyssey into Full Stack Mastery – **Awarded Distinction Certificate.** [PESU-IO]
- OS Fundamentals and Linux – **Awarded Distinction Certificate.** [PESU-IO]

EXPERIENCE

Summer Intern - *Centre of Data Modelling, Analytics and Visualization (CoDMAV), PES University* [July 2025 - August 2025]

CuraMate

[Tech Stack: React.js, FastAPI, PyTorch + scikit-learn (ML), Gemini (LLM)]

- Developed an AI-powered doctor portal to assist with medical triage, symptom clarification, and digital record management during patient consultations.
- Implemented PDF-based patient report parsing to extract demographic details and medical history, reducing manual data entry for doctors.
- Built ML-driven symptom analysis modules to categorize patient conditions and support faster clinical decision-making.
- Integrated LLM-powered clinical explanations and conversational assistance to provide doctors with concise, contextual insights.
- Designed backend APIs using FastAPI to store and retrieve patient records, enabling end-to-end digital workflows from appointment to prescription tracking.

SKILLS

Programming Languages	:	Python, Java, C, JavaScript
Databases	:	SQL (MySQL), Neo4j, Redis
Technologies	:	Linux (Arch, Debian), SSH, REST APIs
Frameworks & Libraries	:	Spring Boot, React.js, Next.js, Node.js, Express.js
Tools	:	Git, GitHub, Vim, Maven, Docker, Postman, Jira
AI Tools	:	Gemini, Claude, ChatGPT (used for development workflows)

PROJECTS

1. Fortress: A Finance Data Backend Application with Role Based Access Control

[Tech Stack: Java, Spring Boot, Maven, SQLite, Postman (API testing)]

- Designed and implemented a scalable backend system for financial data processing with role-based access control using Spring Boot and SQLite.
- Architected a layered backend (Controller–Service–Repository) ensuring clean separation of concerns and maintainability
- Implemented RESTful APIs for transaction lifecycle management including filtering, partial updates, and categorization
- Developed role-based access control (ADMIN, ANALYST, VIEWER) enforcing authorization at the service layer
- Built analytics features including income/expense aggregation, category breakdown, and monthly trends
- Designed robust exception handling and input validation with standardized HTTP responses
- Secured user credentials using SHA-256 hashing and implemented basic authentication workflows

2. Efficient Knowledge Graph Retrieval for LLMs using Graph Partitioning and Hybrid Semantic Search

[Tech Stack: Python, Neo4j, PyTorch, Sentence Transformers, FAISS, NetworkX]

- Designing an efficient KG-RAG pipeline to improve LLM reasoning by combining semantic graph partitioning, ANN-based retrieval, and GNN inference.
- Analyzing limitations of global GNN training on large knowledge graphs and proposing subgraph-level retrieval to reduce latency and noise.
- Conducted an extensive literature survey on Louvain, Leiden partitioning, KG-RAG, GNN-RAG, and hybrid semantic search techniques.
- Evaluated large-scale knowledge graphs (Wikidata-5M, DBpedia, ConceptNet) and justified dataset selection based on semantic richness and scalability.
- Currently implementing a hybrid retrieval pipeline integrating graph embeddings, ANN search, and pre-trained GNN + LLM models.

3. Homeserver: A Personal File Server on a Local Network

[Tech Stack: Java, Spring Boot, Maven, Docker, Watchtower, EndeavourOS, Linux]

- Designed and implemented a layered REST API backend (Controller–Service–Repository) for file storage and retrieval over a local network using Spring Boot.
- Containerized the application using Docker and automated deployments via Watchtower, enabling zero-touch updates by pushing new images to Docker Hub from the dev machine.
- Configured the server laptop to boot into TTY (no GUI) to minimize resource consumption, and set up SSH for remote access and file transfers via SCP.
- Set up systemd to auto-start Docker on boot, ensuring the server and Watchtower containers restart automatically after power outages.
- Exposed the server exclusively on the local network with no public internet exposure, with Tailscale planned for secure remote access.

ACHIEVEMENTS & HACKATHONS

- Took part in **Hacknight 7 (2025)**, a hackathon held by **ACM PESUECC** in which **100+ students took part** and **placed 6th in the leaderboard**.
- **Grabbed the 2nd prize at Heal-O-Code 2.0**, an overnight hackathon held on 28th and 29th March 2025 by the **Centre for Data Modelling, Analytics and Visualization (CoDMAV) & WEAL of PES University, Electronic City Campus**.
- Took part in **Override 23'**, a hackathon held by **Google Developer Student Club (GDSC)** & **placed in the top 10 teams**.

PERSONAL DETAILS

Date of Birth : 25th August 2005
Languages Known : English, Kannada, Telugu and Hindi
Hobbies & Interest : Music (Vocals), Motorsports